Chan Park

AI Engineer | Vision-Language-Action Models for Driving and Robotics

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Summary M.S. in AI (SKKU, 2026): VLA safety repair (first-author RSS 2026 workshop paper, LG Electronics Toronto AI Lab), driving-VLA analysis in CARLA, and deepfake detectors deployed to two Korean law-enforcement agencies.

Education

Sungkyunkwan University Mar. 2024 – Aug. 2026
M.S. in Artificial Intelligence; DASH Lab; Advisor: Prof. Simon S. Woo.

University of Toronto Jan. 2026 – Jul. 2026
Visiting graduate student via CARTE, in partnership with LG Electronics Toronto AI Lab.
Mechanical and Industrial Engineering; Advisor: Prof. Steve Mann.

Inha University Mar. 2017 – Aug. 2023
B.S. in Industrial Engineering.

Experience

LG Electronics, Toronto AI Lab Jan. 2026 – Aug. 2026
AI Research Intern, Robot Learning / VLA Safety (first-author paper, RSS 2026 Workshop) [Project Page]

- Raised SafeLIBERO collision avoidance 64.7%→69.1% and SafeSuccess 47.2%→49.7% over the AEGIS **CBF-QP** filter (4-suite average) with blocked-path detection and entry–exit–rejoin soft guidance.
- Built MuJoCo / RoboCasa simulations, expanded VR demonstrations with MimicGen, and LoRA fine-tuned OpenVLA, π0.5, DreamVLA, and DD-VLA with multi-GPU Slurm training.

DASH Lab, Sungkyunkwan University Mar. 2024 – Dec. 2025
Graduate Researcher, Multimodal Learning / Deepfake Detection [GitHub]

- Maintained $\geq$92.9% video accuracy across all 8 DF-TCB temporal corruptions on FF++ (97.9% clean) with ICR-Net's frame-reliability prediction and selective correction; built DF-TCB on FaceForensics++ and DFDC.
- Led working-level execution of IITP deepfake R&D; deployed Dockerized detection REST APIs to the Korean National Police Agency and Supreme Prosecutors' Office.

Voice AI Lab, Inha University Aug. 2022 – Aug. 2023
Undergraduate Research Assistant

- Built synchronized EMG / EEG / audio / video acquisition for silent-speech research (EMG 1,000 Hz; audio 16 kHz).

Projects

Interpretable Autonomous Driving VLA through Belief Distillation (IBD) Sep. 2026 – Present
Ongoing project | SimLingo / CARLA

- Audited all 2,085,459 auto-generated commentary records: 5.48% of cause-bearing records cite a cause that the generator's own LiDAR visibility flag marks as not visible (up to 38.1% in signalized-junction right turns).
- Showed commentary acts as a control input (**open-loop**, one scenario type): swapping only the cause clause shifted predicted speed by +0.85 m/s (95% CI [0.69, 1.02]); deleting unsupported sentences lost caution in 40.7% of hidden-hazard braking frames vs. 2.5% for hedging.
- Designing IBD: honest commentary plus a camera-only belief token with a readable risk map.

LLM Agent Research Pipeline with Cross-Model Review Sep. 2026
Personal project

- Built a Python cross-model-family reviewer harness on the **open-source ARIS** skill pack for Claude Code: multi-model fallback with back-off, provenance headers, and Codex CLI as second reviewer.
- Flash-tier Gemini scored my two proposals 8.2 / 7.63 vs. code-grounded Codex 5.35 / 5.65 (all: revise); provenance showed the 8.2 came from a fallback model; kept objections open; caught a reviewer error ($\geq$2048 vs. 896).

Submersivity: XR-Guided Underwater Debris Detection Jan. 2026 – Jul. 2026
Mann Lab, University of Toronto [Project Page]

- Raised TACO validation mAP50 from 0.27 to 0.49 by moving from multi-class YOLOv8n to single-class YOLOv8s; reached 92.1% precision (combined validation set) after fine-tuning on 12,177 terrestrial and underwater images.
- Integrated submarine video, YOLOv8s, COLMAP + ArUco, and Vuzix Smart Swim guidance; built the Android XR receiver with packet reassembly, 500 ms stale-data detection, and reconnection.

MetaForensic | KIISE Deepfake Competition, Red-Team Track (team project) Jun. 2025

- Co-built a lip-synthesis pipeline for 436 of 590 silent videos: visual speech recognition (Auto-AVSR), LLM word flips, Kokoro TTS, and MuseTalk lip sync, swapping only the altered-word frames located by Montreal Forced Aligner.
- Sent the 154 videos with unreliable transcripts to an identity-swap branch (ArcFace frame selection + SimSwap, 1–3 spliced segments of 15–45 frames); earned the Excellence Award (Jul. 2025).

Undergraduate Projects

- Built the **3D Figure Generation Platform** with PIFuHD as PM/back-end developer. [Project Page] Apr. 2023
- Found 59 of 168 rest areas suitable for **hydrogen stations** (supervised learning); fine-tuned MobileNet for a **Korean food classifier** app. 2022
- Finished the **Autonomous RC-Car Competition** with the lightest completing car. Fall 2017

Publications

Peer-reviewed (* equal contribution)

1. **Chan Park**, Sunwoo Hong, Kirak Kim, Yoonjeong Park, Alexander W. Olson, and Ali Pesaranghader. "Route-Guided CBF-QP Repair for Safe Vision-Language-Action Models." Workshop on Trustworthy Embodied Foundation Models, Robotics: Science and Systems (RSS) 2026. [Project Page] [Code]
2. Chris McGale, Michel A. Herrera Viyella, **Chan Park**, Daniel Bros, Alexander Vicol, and Steve Mann. "Submersivity: XR-Guided Garbage Detection with an RC Submarine for Peter Street Basin Cleanup." IEEE International Conference on Mechatronics and Automation (ICMA) 2026. [Project Page]
3. Alexander Vicol, Steve Mann, Chris McGale, Gianella Bejar-Alvarez, Michel Herrera Viyella, Darya Zanjanpour, **Chan Park**, Xiaoming Chen, Xueqi Yang, and Bingxuan Yang. "State-of-Flight: Wearable EEG and Motion Context During Flying or Floating." IEEE International Symposium on Computer-Based Medical Systems (CBMS) 2026.
4. **Chan Park**, Hyeongjun Choi, Muhammad Shahid Muneer, Binh Minh Le, and Simon S. Woo. "ICR-Net: Robust Deepfake Detection under Temporal Corruption." Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD) 2026. [Paper] [Code]
5. **Chan Park**, Muhammad Shahid Muneer, and Simon S. Woo. "Beyond Masking: Landmark-based Representation Learning and Knowledge-Distillation for Audio-Visual Deepfake Detection." ACM International Conference on Information and Knowledge Management (CIKM) 2025, Short Paper. [Paper] [Code]
6. **Chan Park***, Bohyun Moon*, Minsun Jeon, Jee-weon Jung, and Simon S. Woo. "X3A: Efficient Multimodal Deepfake Detection with Score-Level Fusion." ACM/SIGAPP Symposium on Applied Computing (SAC) 2025. [Paper]

Manuscripts (not yet peer-reviewed)

7. **Chan Park**. "Interpretable Autonomous Driving VLA through Belief Distillation (IBD)." Ongoing project.
8. "Weighted Contrastive Learning for Robust Audio-Visual Deepfake Detection." Under review. [Code]
9. "Where Are the Artifacts? Residual-Guided Representation Learning for Deepfake Detection and Localization." Under review.

Patents & Software

Method and Apparatus for Deepfake Detection under Temporal Corruption (Korean patent application filed) 2026

Efficient Multimodal Deepfake Detection with Score-Level Fusion (Korean software copyright registration) 2025

Honors & Awards

Korean Government Scholarship | IITP AI Excellence Global Innovative Leader Education Program (MSIT) 2026

Top 10% | 1M-Deepfakes Detection Challenge, ACM Multimedia 2025 [Challenge] 2025

Excellence Award | KIISE Deepfake Generation & Detection Competition (Team MetaForensic) Jul. 2025

SKKU AI Research Fellowship; BK21 Master's Research Fellowship | Sungkyunkwan University 2024–2026

Undergraduate Research Fellowship ($\times$2), Inha University; City-funded Future Talent Scholarship, Siheung City.

Skills

Programming / Systems: Python, C++, Java (Android), SQL; Docker, REST APIs (Flask), Linux, Git, cloud GPU VMs.
ML / LLM: PyTorch, multi-GPU training (Slurm), OpenCV, Hugging Face; Claude Code, Codex CLI, Gemini API, MCP.
Simulation / Robotics: Isaac Sim, MuJoCo, RoboCasa, MimicGen, VR teleoperation.
Languages: Korean (native); English (Advanced).